

§3-5 The Chain Rule

Homework: 3,13,21,23,31,37,39,41,53,57,64,80

The chain rule :

$$(I) (i) \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$(ii) \left\{ f(g(x)) \right\}' = f'(g(x))g'(x).$$

Key to apply the chain rule: Identify u .

註：The chain rule 的証明較 tricky, 可參考課本 P223 的証明。

Example 1 : $y = e^{\sin x}$

(i) Let $u = \sin x$,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = e^u \cdot \cos x = e^{\sin x} \cos x.$$

(ii) $f(x) = e^x$, $g(x) = \sin x \Rightarrow f(g(x)) = e^{\sin x}$

$$\left\{ f(g(x)) \right\}' = f'(g(x))g'(x) = e^{\sin x} \cos x.$$

Example 2 : $y = \sqrt{x + \sqrt{x}}$, Find y' .

Solution : Let $u = x + \sqrt{x}$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{2} u^{-\frac{1}{2}} \left(1 + \frac{1}{2} x^{-\frac{1}{2}} \right) \\ &= \frac{1}{2} (x + \sqrt{x})^{-\frac{1}{2}} \left(1 + \frac{1}{2} x^{-\frac{1}{2}} \right). \end{aligned}$$

Example 3 : $y = \tan(\sin e^{x^2})$, 求 y' .

(Need to apply the chain rule three times)

Solution : Let $u = \sin e^{x^2}$, $v = e^{x^2}$, $s = x^2$.

$$\begin{aligned}
\frac{dy}{dx} &= \sec^2 u \frac{du}{dx} = \sec^2 u \frac{du}{dv} \frac{dv}{dx} \\
&= \sec^2 u \cos v \frac{dv}{dx} \\
&= \sec^2 u \cos v \frac{dv}{ds} \frac{ds}{dx} \\
&= (\sec^2 u)(\cos v)(e^s) 2x.
\end{aligned}$$

Example 4 : $y = \sqrt{\cot(e^{\sin x})}.$

Solution : $\frac{dy}{dx} = \frac{1}{2}(\cot e^{\sin x})^{-\frac{1}{2}}(-\csc^2 e^{\sin x})e^{\sin x} \cos x.$

Example 5 : $y = e^{x \ln a} \left(= e^{\ln a^x} = a^x \right), \quad a > 0.$

Solution : $\frac{dy}{dx} = \ln a \, e^{x \ln a} = (\ln a) a^x$

公式： $(a^x)' = (\ln a) a^x, \quad a > 0.$

Example 6 : $y = |x| = \sqrt{x^2}.$

Solution : $\frac{dy}{dx} = \frac{1}{2}(x^2)^{-\frac{1}{2}}(2x) = \frac{x}{|x|}, \quad x \neq 0.$

$$\Rightarrow \frac{dy}{dx} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}.$$

註：y 在 x=0 的導數不存在。

Example 7 : $y = |\sin x|.$

Solution : $\frac{dy}{dx} = \frac{\sin x}{|\sin x|} \cos x = \begin{cases} \cos x, & \sin x > 0 \\ -\cos x, & \sin x < 0 \end{cases}.$